Simple and Effective VAE Training with Calibrated Decoders

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BAIR
BERKELEY ARTIFICIAL INTELLIGENCE RESEARCH

Calibrated Decoders

Optimal σ-VAE

Gaussian VAE (β=1)

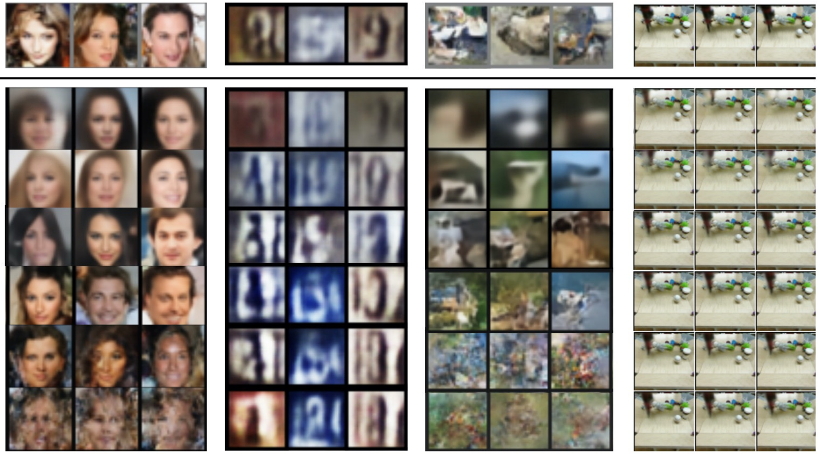
β-VAE, β=10⁻¹

β-VAE, β=10⁻²

β-VAE, β=10⁻³

β-VAE, β=10⁻⁴

β-VAE, β=10⁻⁵



CelebA SVHN CIFAR BAIR

Many works use heuristic β-VAE objective:

$$\mathrm{MSE}(\hat{x}, x) + \beta \mathrm{KL}(q(z \mid x) \| p(z))$$

An equivalent principled objective from ELBO:

$$D \ln \sigma + \frac{D}{2\sigma^2} \mathrm{MSE}(\hat{x}, x) + \mathrm{KL}(q(z \mid x) \| p(z))$$

This σ-VAE learns the β automatically with ELBO

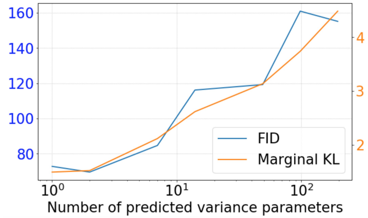
$$\sigma = \sqrt{\beta / 2}$$

Summary:

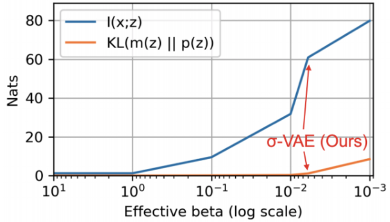
- Statistically calibrated decoders essential for VAEs
- Can learn beta automatically
- Analytic estimate improves performance

Empirical Analysis

Variance Expressiveness

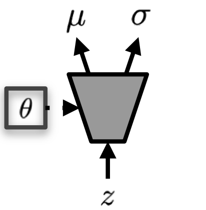


Mutual Information

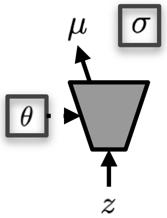


Architectures for Calibrated Decoders

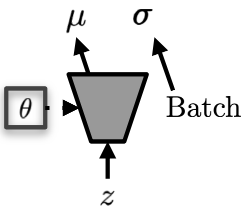
Per-pixel σ-VAE



Shared σ-VAE



Optimal σ-VAE



Image/Video Quality Results

Comparison to Per-pixel σ-VAE

	FID ↓
Gaussian VAE	59.9
β-VAE	53.23
Per-pixel σ-VAE	132.5

Empirical results

	CelebA VAE		SVHN VAE		CIFAR VAE		BAIR SVG [11]	
	$-\log p$ ↓	FID ↓	$-\log p$ ↓	FID ↓	$-\log p$ ↓	FID ↓	$-\log p$ ↓	FID ↓
Bernoulli VAE [14]		177.6		43.26		284.5		122.6
Categorical VAE	< **6359**	71.5	< 9179	46.13	< **7179**	**101.7**	N/A	N/A
bitwise-categorical VAE	< 9067	66.61	< 10800	33.84	< 9390	**91.2**	< 48744	46.13
Logistic mixture VAE	< 7932	65.3	< **9085**	43.19	< 8443	143.1	< **40616**	42.94
Gaussian VAE	< 7173	186.5	< 2184	112.5	< 7186	293.7	< −10379	35.64
β-VAE [23]	< −2713	**61.6**	< −3186	27.93	< −331	**103**	< −13472	34.64
Shared σ-VAE (Ours)	< −6374	**60.7**	< **−3349**	**22.25**	< −5435	116.1	< −13974	34.24
Optimal σ-VAE (Ours)	< **−8446**	**60.3**	< **−3333**	27.25	< **−5677**	**101.4**	< **−14173**	34.13
Opt. per-image σ-VAE (Ours)		66.01		26.28		**104.0**		**33.21**